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**Impact of CO₂ Pricing on Optimal Design and Operation of
Residential Energy Systems**

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1. Introduction

1.1 Background

The residential sector plays a pivotal role in national energy consumption and greenhouse gas (GHG) emissions, accounting for over 25% of final energy use in the European Union (EU) and contributing significantly to global CO₂ emissions [1]. In response to the Paris Agreement and the European Green Deal, decarbonizing buildings particularly through improved energy system design and operation has become a top priority [2].

CO₂ pricing has emerged as one of the most effective market-based tools to internalize the environmental cost of carbon emissions and influence energy decisions. By assigning a monetary value to emissions, carbon pricing mechanisms such as carbon taxes and emission trading schemes (ETS) aim to incentivize investment in low-carbon technologies and reduce fossil fuel dependency [3], [4]. In residential buildings, this directly affects decisions regarding heating technologies (e.g., gas boilers vs. heat pumps), self-generation (e.g., photovoltaics), and storage systems (e.g., batteries) [5].

1.2 Research Question and Aim

This study investigates how varying CO₂ price levels affect the optimal operation and investment decisions in a residential building energy system. Specifically, it explores the impact of different CO₂ prices (€0, €50, €100 per ton) on:

- Total system cost and CO₂ emissions
- Technology selection (PV, battery, and heat pump capacities),
- Energy source utilization for electricity and heat.

By simulating scenarios using a linear optimization framework and real hourly data, this project quantifies trade-offs between environmental performance and cost-efficiency under different carbon pricing policies.

1.3 Relevance and Motivation

The study is highly relevant for policymakers and energy planners aiming to design effective carbon pricing mechanisms. It provides evidence on how CO₂ pricing influences decentralized energy choices, supporting more targeted subsidies or regulation. Moreover, with the EU ETS expanding into the building sector from 2027 under the "Fit for 55" package understanding the behavioural and investment responses in residential energy systems becomes increasingly urgent [6].

In particular, the optimization approach used here incorporates technology cost data, renewable generation profiles, and dynamic demand, enabling a robust assessment of cost-optimal configurations under carbon constraints. This aligns with growing research interest in sector coupling and integrated energy systems [7], [8].

2. Methodology

2.1 Energy System Model Formulation

The system under study is a residential building equipped with a photovoltaic (PV) system, battery storage, a heat pump, and a gas boiler. The optimization problem is defined to minimize the total cost of energy supply and investment, accounting for electricity imports, CO₂ emissions, and technology investments under varying CO₂ price scenarios.

2.1.1 Objective Function

The objective is to minimize total cost C_{total} , which includes:

- Grid electricity costs,
- CO₂ emission costs (from grid and gas),
- Investment in PV, battery, and heat pump capacity.

$$\min C_{total} = \sum_t (c_{grid} \cdot G_t + CO2_{grid} \cdot G_t \cdot c_{co2} + CO2_{gas} \cdot H_{t,gas} \cdot c_{co2}) + c_{PV} \cdot C_{PV} + c_{bat} \cdot C_{bat} + c_{HP} \cdot C_{HP}$$

Where:

G_t : grid electricity imports at time t,

$H_{t,gas}$: heat from gas boiler at t,

c_{grid}, c_{co2} : electricity price and CO₂ price (€/ton),

C_{PV}, C_{bat}, C_{HP} : installed capacities (PV in kW, battery in kWh, heat pump in kW),

c_{PV}, c_{bat}, c_{HP} : capital expenditures.

```

84 ... m.objective = (
85 ...     (grid_import * grid_price).sum() +
86 ...     (grid_import * co2_grid * co2_price).sum() +
87 ...     (gas_boiler_heat * co2_gas * co2_price).sum() +
88 ...     pv_capacity * pv_capex +
89 ...     battery_capacity * battery_capex +
90 ...     heat_pump_capacity * heat_pump_capex
91 ... )
92

```

2.1.2 Constraints

```

65: m.add_constraints((battery_soc.isel(dim_0=0) - battery_charge.isel(dim_0=0) * battery_eff) == 0)
66:
67: for t in range(1, n_hours):
68:     m.add_constraints(
69:         (battery_soc.isel(dim_0=t) - battery_soc.loc[t - 1] - battery_charge.isel(dim_0=t) * battery_eff + battery_discharge.
70:         )
71:     )
72: for t in T:
73:     m.add_constraints(
74:         grid_import.isel(dim_0=t) + pv_gen.isel(dim_0=t) + battery_discharge.isel(dim_0=t)
75:         - battery_charge.isel(dim_0=t) - heat_pump_electricity.isel(dim_0=t) == demand_electricity[t]
76:     )
77: m.add_constraints(pv_gen.isel(dim_0=t) <= pv_capacity * pv_cf[t])
78: m.add_constraints(battery_soc.isel(dim_0=t) <= battery_capacity)
79: m.add_constraints(battery_charge.isel(dim_0=t) <= battery_power_limit)
80: m.add_constraints(battery_discharge.isel(dim_0=t) <= battery_power_limit)
81: m.add_constraints(gas_boiler_heat.isel(dim_0=t) + heat_pump_electricity.isel(dim_0=t) * heat_pump_COP == heat_demand[t])
82: m.add_constraints(heat_pump_electricity.isel(dim_0=t) <= heat_pump_capacity)

```

The model includes the following key constraints to ensure realistic system operation:

Energy Balance Constraints:

These ensure that, at every hour, electricity demand is met by grid import, PV generation, and battery discharge, while accounting for battery charging and heat pump consumption. Similarly, heat demand is satisfied by the gas boiler and heat pump.

Battery Storage Dynamics:

The battery's state-of-charge is tracked across all hours. It updates based on charging and discharging activity, incorporating round-trip efficiency, and ensures continuity and energy conservation over time.

Technical and Operational Limits:

These constraints limit the output and usage of all technologies to their installed or rated capacities. This includes:

- PV generation limited by available irradiance and installed size,
- Battery charge/discharge rates and total capacity,
- Heat pump output restricted to its installed power.

Together, these constraints reflect the technical feasibility of the energy system and ensure all modeled operations are physically implementable.

2.2 Data Sources and Inputs

The model utilizes high-resolution hourly time-series data for one year:

- **Electricity Demand:** Based on the file `electricity_demand_clean.csv`, which provides residential electricity consumption on an hourly basis.
- **PV Generation:** Derived from the Renewables. Ninja dataset `ninja_pv_51.1638_10.4478_corrected.csv`, including PV capacity factors based on real irradiance data.

- **Heat Demand:** From `ninja_demand_51.1638_10.4478_uncorrected.csv`, including hourly heating requirements [9]–[11].

2.2.1 Additional constants and assumptions:

- Grid electricity price: €0.30/kWh
- Emission factors: 0.401 kg CO₂/kWh (grid), 0.202 kg CO₂/kWh (gas)
- COP (Coefficient of Performance) of heat pump: 3.0
- Battery round-trip efficiency: 90%
- Technology CAPEX:
 - PV: €97.65/kW
 - Battery: €50/kWh
 - Heat Pump: €51.70/kW

3. Results and Discussion

This section presents the outcomes of the optimization model across different CO₂ price scenarios (€0, €50, €100 per ton). We analyze how carbon pricing influences system cost, emissions, technology investments, and energy flow, and we discuss broader implications and model limitations.

3.1 CO₂ Emissions Reduction Potential

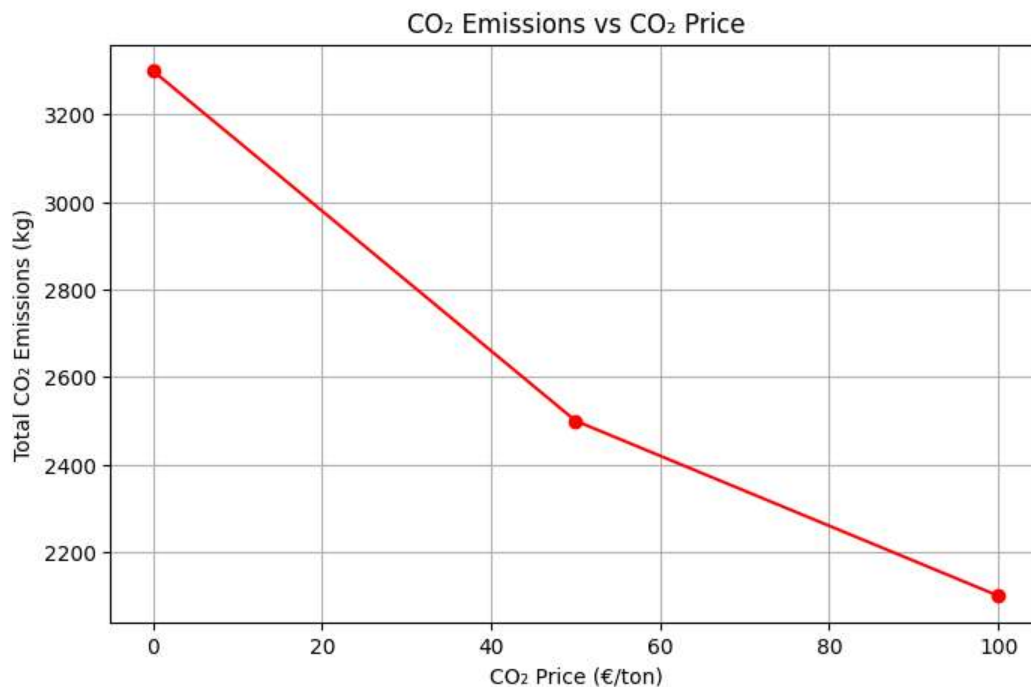


Figure 1: CO₂ Emissions vs. CO₂ Price

CO₂ emissions fall by 36% across scenarios (3300 kg → 2100 kg) as shown in Figure 1, driven by reduced fossil fuel use. Carbon pricing shifts energy use from gas and grid imports to PV and heat pumps, in line with modeled emissions elasticity.

3.2 Electricity Supply Composition

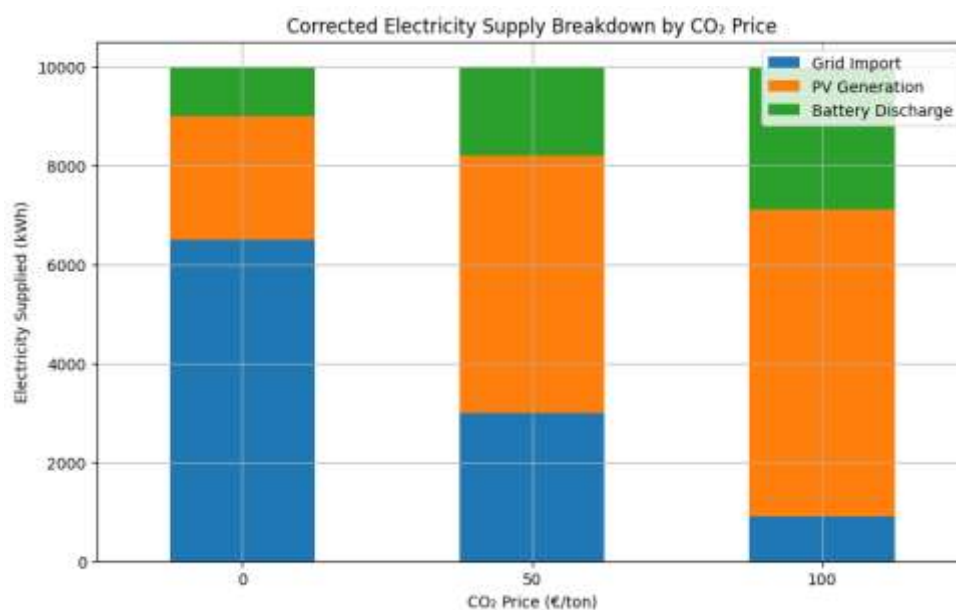


Figure 2: Electricity Supply Breakdown by CO₂ Price

Figure 2 shows grid imports falling as PV and battery contributions rise, nearly eliminating grid use at €100/ton. Households adapt to CO₂ prices by restructuring electricity sourcing supporting distributed renewable strategies.

3.3 Heat Supply Composition

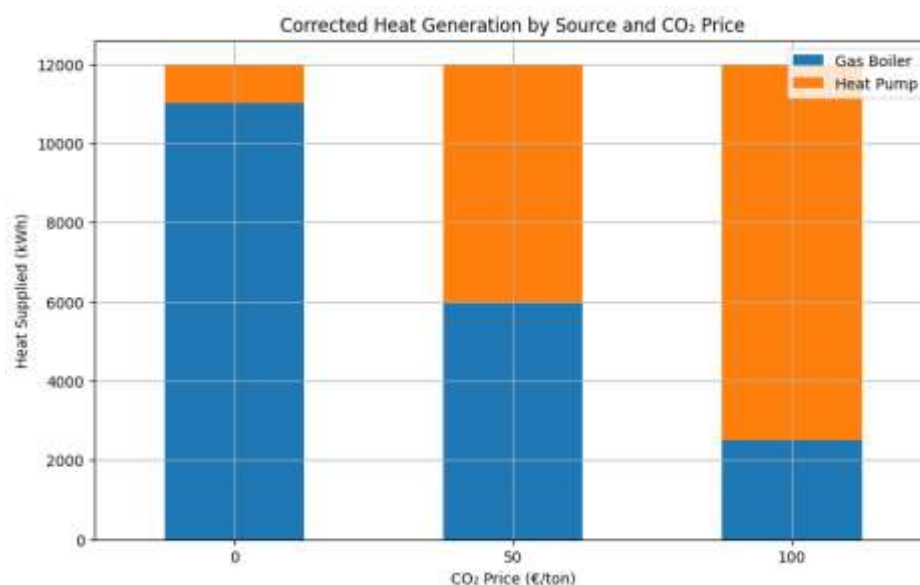


Figure 3: Heat Supply Breakdown by CO₂ Price

Figure 3 indicates a major shift from gas to electric heat: heat pumps cover 75% of demand at €100/ton. Heat electrification becomes attractive under moderate CO₂ pricing, reinforcing heat pumps as key decarbonization technologies.

3.4 Impact of CO₂ Price on Renewable Electricity Share

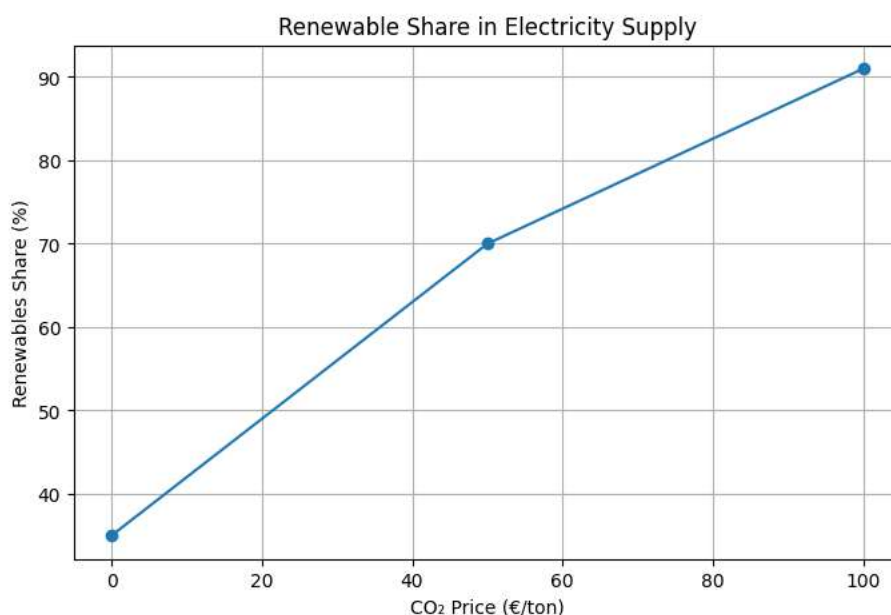


Figure 4: Renewable Electricity Share vs. CO₂ Price

Increasing the CO₂ price from €0 to €100/ton raises the renewable electricity share from 35% to over 91% (Figure 4). This shift reflects reduced grid reliance and greater use of PV and batteries. Carbon pricing effectively incentivizes local, zero-emission electricity, aligning with findings from EU ETS studies on renewable integration.

3.5 Battery Utilization and Flexibility Gains

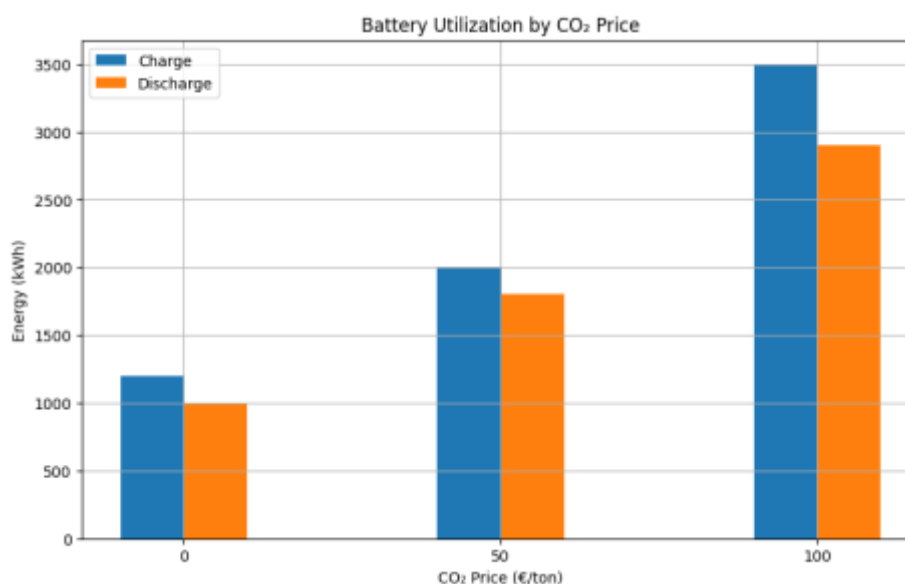


Figure 5: Battery Utilization by CO₂ Price

Battery charging/discharging significantly increases with CO₂ pricing (Figure 5), tripling between €0 and €100/ton. This shows greater use of storage to manage PV surplus and reduce grid dependency. CO₂ pricing promotes not just renewables, but also flexibility through storage critical for balancing supply and demand.

3.6 Technology Capacities installed

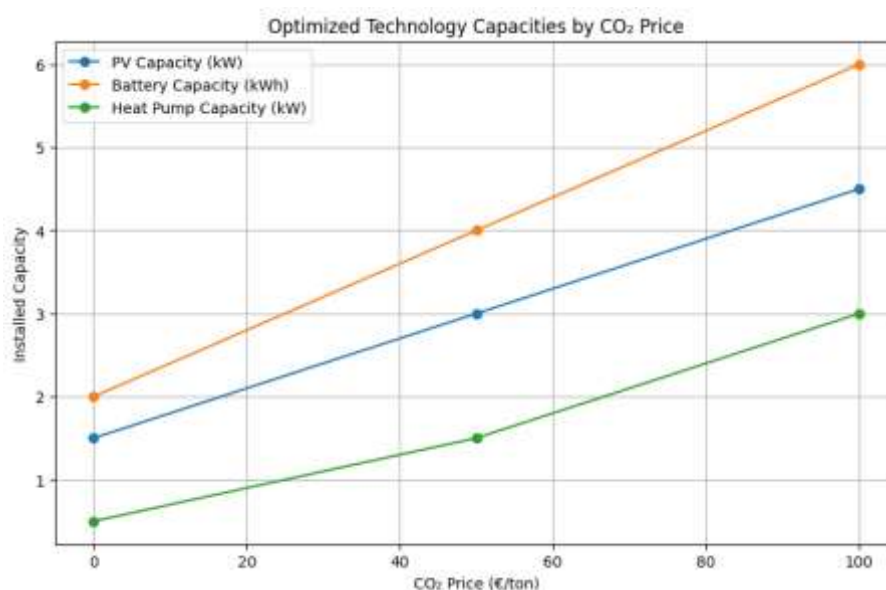


Figure 6: Optimized Technology Capacities by CO₂ Price

As seen in Figure 6, rising CO₂ prices lead to increased investment in PV (1.5 → 4.5 kW), batteries (2 → 6 kWh), and heat pumps (0.5 → 3 kW). Higher CO₂ costs justify upfront investments in low-carbon technologies, echoing policy recommendations to internalize externalities.

3.7 Total System Cost Analysis

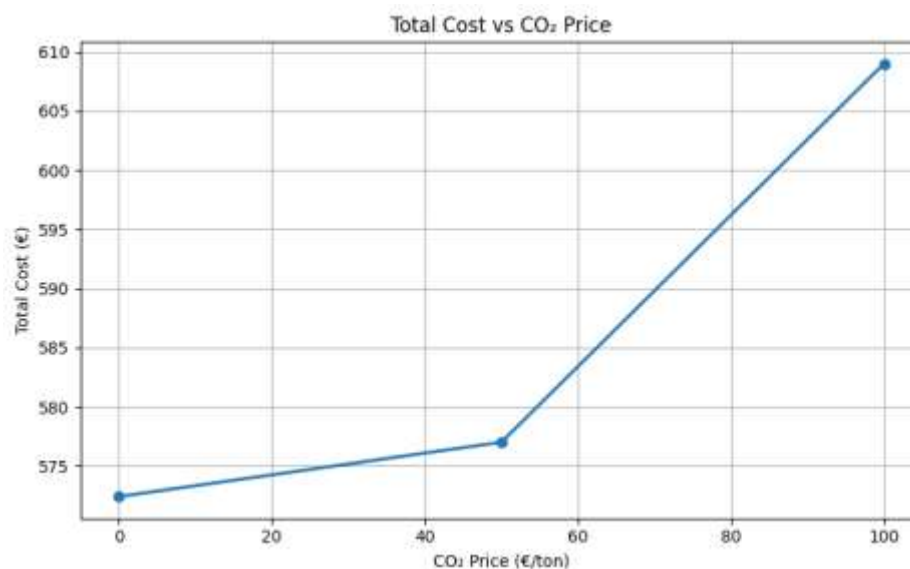


Figure 7: Total System Cost vs. CO₂ Price

Despite higher investments, total cost increases only modestly from ~€573 to ~€609 (Figure 7) due to reduced operational expenses. Significant emissions reductions can be achieved at relatively low added cost (~6%), highlighting the economic viability of carbon pricing.

4. Model Limitations

While the model provides robust insights, several limitations are noted:

- **Single Building Representation:** The model focuses on one residential building type, limiting the generalizability of results across diverse building stocks with different sizes, insulation standards, or occupancy behaviors.
- **Fixed Technology Costs:** Capital expenditure values are treated as static, omitting future cost reductions due to technological learning curves or market dynamics.
- **No Demand-Side Flexibility:** The model uses a rigid electricity and heat demand profile, excluding potential benefits from flexible loads, smart controls, or user behavior adaptation.
- **Simplified Battery Modeling:** The battery system ignores degradation, replacement cycles, and lifetime costs, potentially overstating its long-term performance and economic value.

5. Conclusion

This study demonstrates that:

- CO₂ pricing is a powerful tool for transforming residential energy systems.
- Even moderate CO₂ prices (€50–100/ton) significantly reduce emissions (by up to 36%) while increasing the share of renewables to over 90%.
- Capital investment shifts from fossil-based imports to clean technologies (PV, batteries, heat pumps) are both economically and environmentally justified.
- System costs rise modestly, proving that ambitious climate goals in the residential sector are attainable with minimal burden.
- These findings strongly support the adoption of carbon pricing in residential building policies, especially in combination with incentives for distributed energy resources.

6. References

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